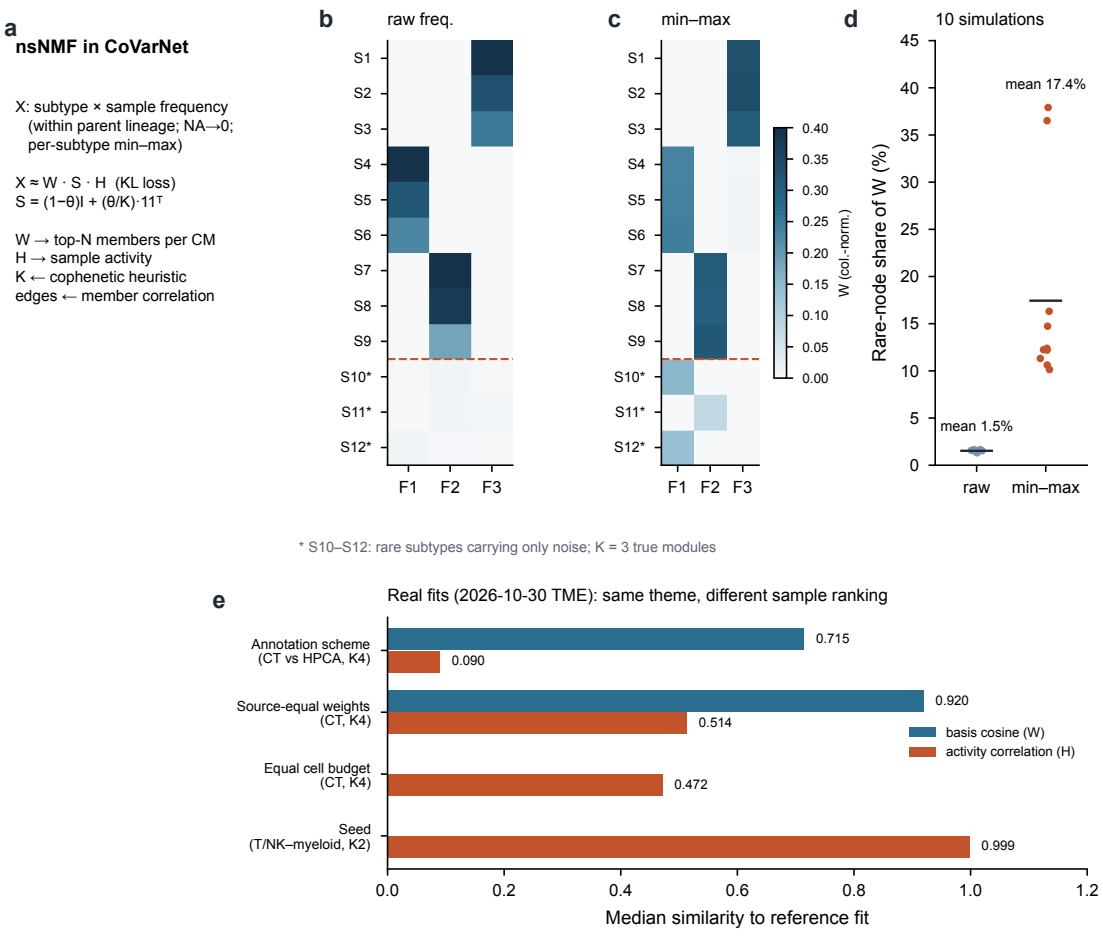
a
nsNMF in CoVarNet
X: subtype × sample frequency
(within parent lineage; NA→0;
per-subtype min–max)
X ≈ W · S · H (KL loss)
S = (1−θ)I + (θ/K)·11ᵀ
W → top-N members per CM
H → sample activity
K ← cophenetic heuristic
edges ← member correlation
b
raw freq.
c
min–max
S1
S2
S3
S4
S5
S6
S7
S8
S9
S10*
S11*
S12*
F1
F2
F3
W (col.-norm.)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
d
10 simulations
Rare-node share of W (%)
mean 17.4%
mean 1.5%
raw
min–max
* S10–S12: rare subtypes carrying only noise; K = 3 true modules
e
Real fits (2026-10-30 TME): same theme, different sample ranking
Annotation scheme
(CT vs HPCA, K4)
0.715
0.090
Source-equal weights
(CT, K4)
0.920
0.514
Equal cell budget
(CT, K4)
0.472
Seed
(T/NK–myeloid, K2)
0.999
basis cosine (W)
activity correlation (H)
Median similarity to reference fit